

Data cleaning in excel

Dataset: FIFA 21 raw data from kaggle

Link: <https://www.kaggle.com/datasets/yagunnersya/fifa-21-messy-raw-dataset-for-cleaning-exploring>

Tools used: Excel

Data quality issue	Column name	how to resolve the issue
Accuracy	w/f - inaccurate; sm - inaccurate; ir- inaccurate	Remove the columns as it doesn't make sense
Completeness	Loan column -blanks ; hits- blanks	Don't remove it see the explanation bellow
Consistency	Value, wage, release cause - inconsistent	Strip the star cases
Readability	A^+ova - overall rating ; pot- potential	Change the column names
Uniqueness	no duplicates	--
Validity	Contract - format; height - format; weight- format; value, wage, release cause- format	convert it to number format
Relevancy	Contract - loan; contract - free (filter out)	make it a separate sheet for any further need

Explanation by columns:

1. SHORT NAME: Remove

As we can see in the raw data the short name column and long name column is there. Either one is enough to know their names so let's delete the short name column

2. A+OVA: Change the column name for better readability

As per [Sofia.com](https://sofia.com) a+ova refers to overall rating. So convert it to that name

3. POT: Change the column name for better readability

As per [Sofia.com](https://sofia.com) pot refers to potential. So convert it

4. CLUB: Certain rows were not visible because it was wrapped inside it so unwrap it.

Click the unwrap text in home ribbon.

5. HIEGHT: Convert it to number format

6. WEIGHT: Convert it to number format

7. CONTRACT: Change the format And certain inaccuracies

- Change the format to date
- Certain values in contract says that they are in loan so no end contract is given in that particular column but it has given a separate column named loan date ended so let's concatenate both joined loan year and end loan year by using concatenation formula.

```
Concatenate(Right(Q207,4)&"~"&Right(R207,4))
```

Convert : june21 2013to 2013~2021

2013~2021

This is how we get this format.

9. VALUE: Strip the star values

Eg: a^,-103.5m to just 103.5m

```
=mid(s2,4,9)      #s2 = a^, -103.5m
```

103.5 M

10: WAGE: Strip the star values

Eg: a^,-560K to just 560K

```
=mid(u2,4,7)      # u2 = a^, -560k
```

560k

11. RELEASE CLAUSE: Strip the star values

Eg: a^,-560K to just 560K

```
= mid(w2,4,9).      #w2 = a^, -138.4m
```

138.4m

12. HITS : Either remove it or look for the data. if the analysis needs hit.

13. LOAN: Here loan column has lots of blanks because most of the data in it is not under loan. so we should not delete it rather we can remove those contract under loan and delete the column or just use it as it wont harm the data.

Feature engineering:

1. To find number of years of contract

Let's create a separate column for start year and end year by using right and left formula

```
=left(j2,4) to find start year # j2 is 2004~2021  
=right(j2,4) to find end year.
```

Start year column : 2004

End year column: 2021

create a separate column and subtract end year from start year to get no of years.

2. Create a age brackets to know how many fall below 25,between 25-39 and above 40.

```
=if(and(f2<25),"below 25",if(and(f2>=25,f2<=39),"25-39",if(and(f2>=40),"above 40"))))
```

25-35

create a separate column and apply the formula and u will get result like this.

3. Create physical points, mental points, technical points, goalkeeping points.

Physical points includes:

Physical points	Mental points	Technical points
• Agility • Acceleration • Balance • Jumping • Reactions • Sprint speed • Stamina • Strength	• Aggression • Interceptions • Positioning • Vision • Composure	• Crossing • Finishing • Heading accuracy • Short passing • Volleys • Dribbling • Curve • Fk accuracy • Long passing • Ball control • Shot power • Long shots • Penalties • Defensive • Standing tackle • Sliding tackle

As per [Sofia.com](#) these are the points included per head. Goalkeeping points is already given in the data.

Created a separate column for each and added these points. And created total points.



cleaned fifa data.xlsx